

NEUROPHARMACOLOGY LAB REPORT — COMBINATION PROTOCOL EXPERIMENT REPORT

TARGET: ORGANISM TYPE: COMBINATION

COMPOUND DETAILS

Compound	Ketamine	Drug Class	Dissociative Anesthetic
Dose (mg eq.)	0.5mg	BBB Coefficient	0.90
Receptor Targets	NMDA, 5-HT2A, GABA-B, D1		
Compound 2	MDMA	Drug Class 2	Antipsychotic
Dose 2 (mg eq.)	10mg	BBB Coefficient 2	1.00
Receptor Targets 2	D2, D1, NMDA		
Session Date	2026-06-16T21:46:56.903Z		

Tests run in an isolated Controlled Lab session. Live brain was not modified. Platform biologically constrained by HCP-MMP1.0, Allen Brain Atlas, and Brainnetome Atlas data.

PEAK RECEPTOR OCCUPANCY

NMDA	5-HT2A	GABA-B	D1	D2
0.1%	0.1%	0.1%	0.1%	0.1%

Occupancy curves follow rapid absorption / first-order elimination kinetics (high-level approximation). Not equivalent to full pharmacokinetic modeling.

REGIONAL ACTIVATION EFFECTS

Region	Receptor	Activation Delta	Predicted Effect (consistent with literature)
Thalamus	D2	-26.9%	increased compulsive motor patterns
AnteriorCingulateCortex	NMDA	-26.9%	impaired memory encoding
Brainstem	GABA-B	-26.9%	reduced thalamocortical relay fidelity
AnteriorCingulateCortex	NMDA	-26.9%	impaired memory encoding
AnteriorCingulateCortex	5-HT2A	-26.9%	suppressed representational diversity
Hippocampus	GABA-B	-26.9%	reduced thalamocortical relay fidelity
NucleusAccumbens	D1	-26.9%	impaired prefrontal gating
NucleusAccumbens	D1	-26.9%	impaired prefrontal gating
BasalGanglia	D1	-26.9%	impaired prefrontal gating
BasalGanglia	D1	-26.9%	impaired prefrontal gating
Insula	5-HT2A	-26.9%	suppressed representational diversity
NucleusAccumbens	D2	-26.9%	increased compulsive motor patterns

Effects represent simulated predictions from population-level receptor binding models. Not equivalent to in vivo pharmacology data.

VALIDATION SUMMARY

Integrity: PASS Stability: PASS Verdict: WEAK EFFECT

ANSInteroceptiveCoupling did not produce a measurable separation from baseline or shuffled control on pre-registered metrics. Claims of biological significance are not supported by this run.

Language note: This report uses "consistent with", "suggests", or "did not support" for all mechanistic claims. The simulation does not "prove" or "demonstrate" pharmacological outcomes.

PHARMACOKINETIC PROFILE

Cmax (normalized)	0.002	Tmax	27.8 ticks
Half-life	36.1 ticks	ke (elimination)	0.0192
AUC	0.09	EC50	0.35
Hill Coefficient	1.72	BBB Penetration	0.90

PK parameters derived from two-compartment rapid-absorption / first-order elimination model. Not equivalent to full clinical PK study.

NEUROCHEMICAL CASCADE MAP

Neurochemical	Net Delta
Glutamate	+0.0%
BDNF	+0.0%
Nitric Oxide	+0.0%
Serotonin	+0.0%
Anandamide	+0.0%
GABA	+0.0%
Beta-Endorphin	+0.0%
Dopamine	+0.0%
Adenosine	+0.0%
serotonin	+0.0%
NPY	+0.0%

From	To	Mechanism	Delta
Glutamate	BDNF	agonist cascade	+0.0%
BDNF	Nitric Oxide	agonist cascade	+0.0%
Serotonin	Glutamate	agonist cascade	+0.0%
Glutamate	Anandamide	agonist cascade	+0.0%
GABA	Beta-Endorphin	agonist cascade	+0.0%
Beta-Endorphin	Serotonin	agonist cascade	+0.0%
Dopamine	BDNF	agonist cascade	+0.0%
BDNF	Adenosine	agonist cascade	+0.0%
Dopamine	serotonin	agonist cascade	+0.0%
serotonin	NPY	agonist cascade	+0.0%
Dopamine	BDNF	agonist cascade	+0.0%

BDNF	Adenosine	agonist cascade	+0.0%
Glutamate	BDNF	agonist cascade	+0.0%
BDNF	Nitric Oxide	agonist cascade	+0.0%

Neurochemical deltas are derived from receptor-binding-to-second-messenger cascade models. Cascade sequence represents dominant literature pathway, not exhaustive pharmacology.

TIME-SERIES RESPONSE

Tick	Time (ms)	Plasma Concentration
0	0	0.0000
5	4,365	0.0000
10	8,730	0.0000
15	13,095	0.0000
20	17,460	0.0000
25	21,825	0.0000
30	26,190	0.0010
35	30,555	0.0010
40	34,920	0.0010
45	39,285	0.0010
- ... -		
275	240,075	0.0010
280	244,440	0.0010
285	248,805	0.0010
290	253,170	0.0010
295	257,535	0.0010

First 10 + last 5 time points shown. Full series: 60 ticks at 4365ms/tick intervals.

STATISTICAL VALIDATION

Cohen's d	0.005	Effect Size Class	small
95% CI Lower	-0.683	95% CI Upper	0.687
ESURIENS Hunger-Drive Impact	0.1%	Memory Temple Encoding Δ	30.0%

Statistical parameters estimated from single-session simulation. Cohen's d < 0.2: small, 0.2-0.5: medium, 0.5-0.8: large, > 0.8: very large. Confidence interval assumes gaussian error propagation.